

Part A

Client Brief SmartCare currently relies on spreadsheets and paper records. Staff experience duplicate bookings, difficulty finding patient information, inconsistent appointment statuses and limited appointment history. The proposed solution is a small and maintainable system for managing patient, practitioner and appointment information.

Part B

In Scope

- Create, view, update and search patient records
- Create, view and update practitioner records.
- Create, view, update and cancel appointments
- Link each appointment to a patient and practitioner
- Use consistent appointment statuses
- Detect or prevent duplicate appointment bookings
- Detect or prevent duplicate appointment bookings

Provisional Until Confirmed

- Practitioners directly log in to view their schedules.
- Cancelled appointments are retained indefinitely in history
- Role-based access, audit logging and specific performance targets.

Out of Scope

- Online payments and billing.
- SMS or email reminders
- Facial recognition

StakeHolder	Need	Evidence
Reception Staff	Efficiently find information and create, update, or cancel appointments for patients.	Report states staff problems with booking, information, status, and history.
Practitioners	Schedules and appointment statuses that are accurate.	Brief said practitioners need appointment management
Patient	Personal information, appointments, and history.	Patient information and appointments are managed by the proposed system.
Management	Mainainble system that reduces duplicate bookings and inconsistent records	Stated in brief.
Developers and testers	Testable requirements and maintainable business logic	Necessary delivery stakeholders.

Part C

FR-01: System shall allow authorised staff to create a patient record. Why: supports management of patient information.

FR-02: system allow authorised staff to view and update a patient record. Why: Supports correction and maintenance of patient information.

FR-03: System shall allow authorised staff to search for a patient using an approved identifier or search field. Why: Directly addresses difficulty finding patient information; the exact search fields are an open question.

FR-04: System must allow permitted personnel to create, examine and edit practitioner records

Why: The requested solution must manage practitioners.

FR-05: Allow authorised staff to create an appointment linked to one patient and one practitioner. Why: Defines the central appointment capability and its record relationships.

FR-06: Before the appointment is saved the system looks for a any conflicting appointments. Why: Stops duplicate reservations: Important to clarify whats a dispute.

FR-07: Has to show the patient, doctor, time, date and current status of the appointment. Why: These information are required.

FR-09: The system shall allow authorised staff to cancel an appointment without deleting its record. Why: Supports cancellation while preserving history; the retention rule requires confirmation.

Part D

NFR-01: Maintainability: patient, practitioner and appointment business logic shall be separated from the user interface so it can be changed and tested independently. Why: The brief explicitly asks for a maintainable system.

NFR-02: Data integrity: If an appointment doesn't have a legitimate patient, practitioner, date, time, or approved status, the system will refuse it. Why: Prevents inconsistent or incomplete recordings of appointments.

NFR-03: Reliability: The stored record must be accessible even after the programme has been closed and reopened once the system has verified a save. Why? Because it makes persistence testable and does not presume the necessary storage technology.

NFR-04: Usability: A receptionist must be able to find an existing patient and begin making an appointment on their own after receiving the agreed-upon training; the client must confirm the acceptable time and mistake rate. Why? Because it substitutes a testable task with open-ended objectives for vague "easy to use" terminology

Part E

US-01: As a receptionist, I want to search for a patient, so that I can find the correct patient before making a booking.

US-02: As a receptionist, I want to create an appointment linked to a patient and practitioner,

so that the booking details are recorded consistently.

US-03: As a receptionist, I want the system to warn me about a conflicting booking, so that I can avoid duplicates.

US-04: As a receptionist, I want to update an appointment's status, so that staff can see its current state.

US-05: As a receptionist, I want to cancel an appointment without removing its record, so that the appointment can remain in history if the client approves this rule.

Acceptance Criteria

AC-01 Successful patient search

GIVEN an existing patient record with an approved identifier

WHEN an authorised receptionist searches using that identifier

THEN the system displays the matching patient's information.

AC-02 Successful appointment creation

GIVEN a valid patient, practitioner, date, time and approved status, with no conflicting booking

WHEN an authorised receptionist saves the appointment

THEN the system stores the appointment and displays a confirmation.

AC-05 Appointment cancellation

GIVEN an existing active appointment

WHEN an authorised receptionist cancels it

THEN the system changes its status to Cancelled and does not remove the record, subject to the approved retention rule.

Part F

Review prompt Act as a software requirements reviewer. Review the SmartCare requirements for ambiguity, inconsistency, missing clarification questions and testability. Do NOT invent new client requirements. For every suggestion, state whether it is based on evidence or is only a question/assumption requiring validation.

Part G

Define duplicate booking precisely.

Evidence: Brief reports duplicate bookings.

Decision: Accepted

Reason: We need to stop a conflict to make the requirement testable. Verification: Ask which combinations and time overlaps count as conflicts.

Make search performance measurable.

Evidence: Brief reports difficulty finding patient information.

Decision: Modified

Reason: AI cannot choose the response time or dataset size. Verification: Get client approved data and run a timed test.

Retain cancelled appointments forever.

Evidence: Brief only says appointment history is limited.

Decision: Unverified

Reason: The retention period is not stated. Verification: Get retention and deletion rules with the client.

Add SMS reminders.

Evidence: No supporting evidence.

Decision: Rejected

Reason: Not requested; adds integration scope. Verification: Include only if the client approves a scope change

Add facial recognition login.

Evidence: No supporting evidence.

Decision: Rejected

Reason: Authentication is unspecified; biometrics add complexity and risk. Verification: Ask for security requirements without assuming biometrics.

Assumptions and Open Questions

Who will use the system, and what can each person see or change?

When do appointments clash—for example, the same patient or practitioner booked at the same time?

Should cancelled appointments be kept? If so, for how long?

Does the system need logins, access limits, records of changes, backups or privacy protection?

Should existing spreadsheet and paper records be added to the system?